

Concept of a Random Variable

Statistics is concerned while making inferences about the populations & population characteristics. Experiments are conducted with results that are subject to chance. The testing of a no. of electronic components is an example of statistical experiment, a term that signifies any process by which several chance observations are generated.

A random variable is a function that associates a real number with each element in the sample space.

We generally use capital letter X to denote a random variable, and x to show any of its values.

There are several cases where the random variable is categorical in nature. In such cases, variables called dummy variables are used.

The random variable for which 0 & 1 are chosen to describe the two possible values is called a Bernoulli random variable.

Statisticians often use sampling plans to either accept or reject batches or lots of material.

- If a sample space contains a finite no. of possibilities or an unending sequence with as many elements as there are whole nos., it is called a discrete sample space.
- If a sample space contains an infinite no. of possibilities equal to no. of points on a line segment, it is called continuous sample space.
- A random variable is called a discrete random variable if its set of possible outcomes is countable.
- When a random variable can take on values on a continuous scale, that becomes a continuous random variable.
- Egs. of continuous random variable
 ↳ measured data → ht., wt., temp., distance, etc.
- Egs. of discrete random variable
 ↳ count data → no. of fatalities, no. of defective samples, etc.